

input is predefined method which is used to take input from user through keyboard.

*input () always return string.

Syntax:

input ("Prompt")

*) find avg of 3 numbers.

1. input $\rightarrow$ 3 integers

2. Process $\rightarrow$ logic / condition

3. output $\rightarrow$ Avg.

Algorithm

① start

② take 3 numbers, n_1, n_2, n_3

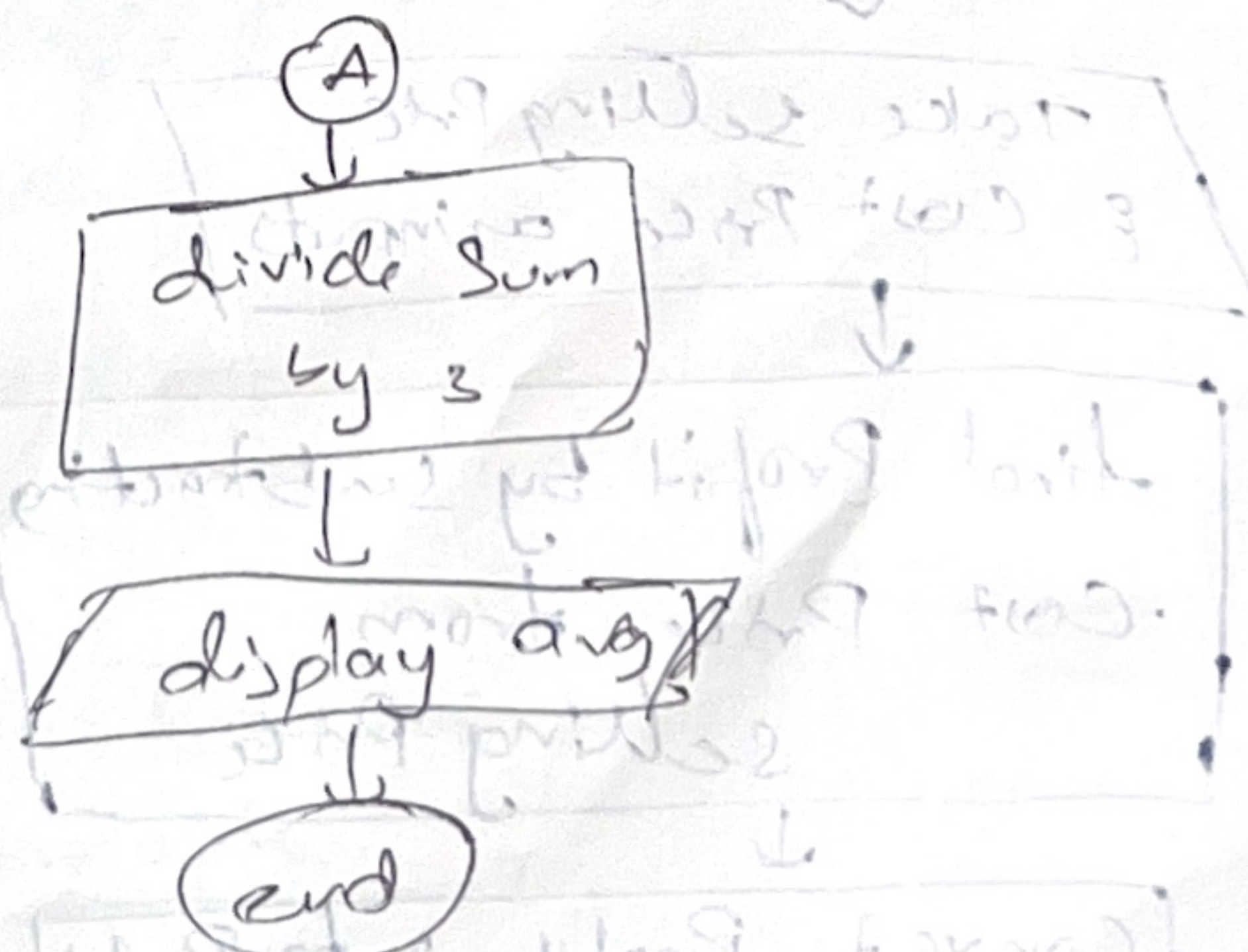
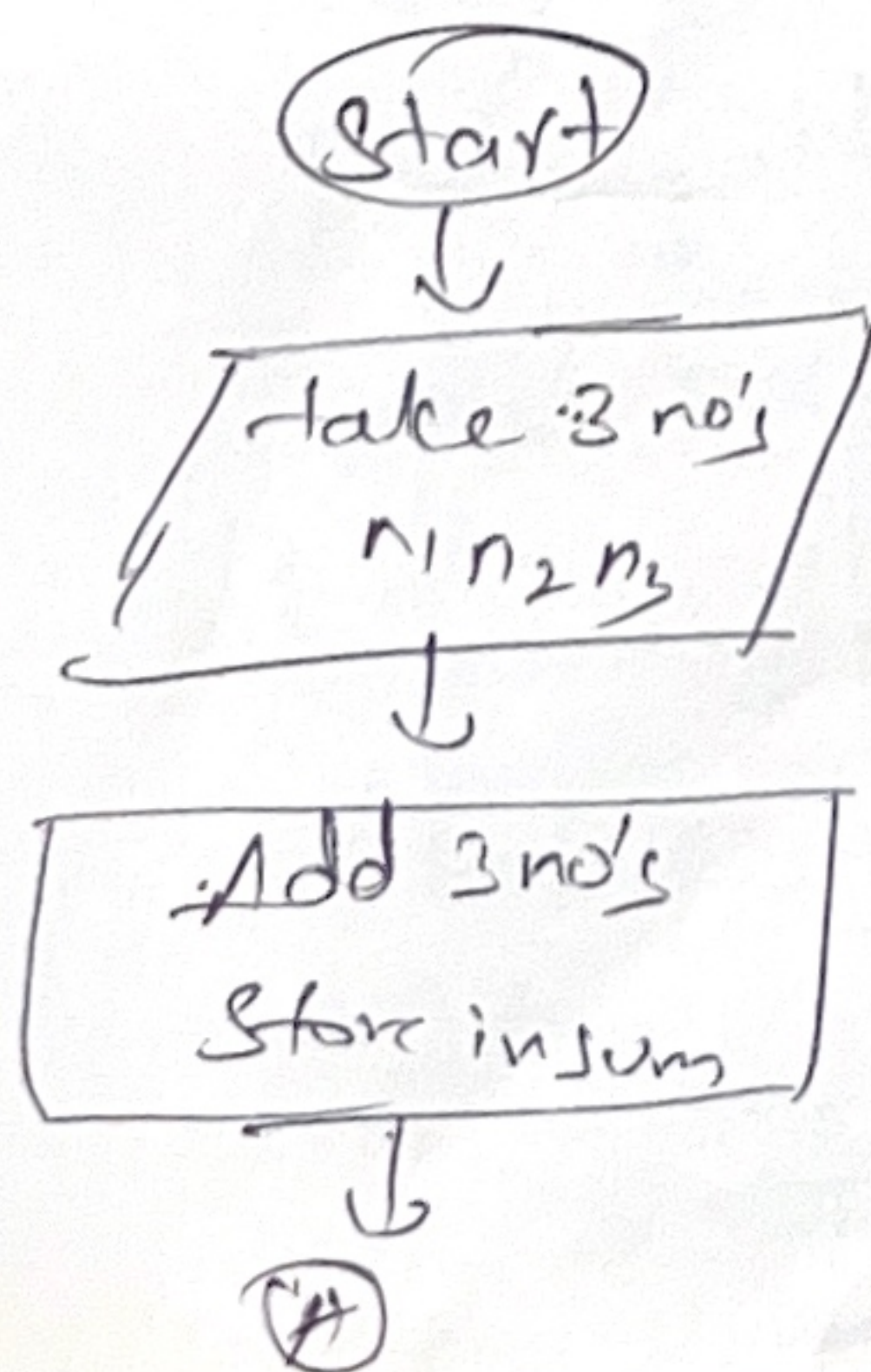
③ Add three numbers and store in sum

④ End or calculate avg of three and display.

⑤ display Avg

⑥ end

flowstart



2. find Profit Percentage by Using selling & cost Price

① input $\rightarrow$ SP, CP

② Process $\rightarrow$ Profit
$$PP = \frac{P}{Cp} \times 100$$

③ output $\rightarrow$ Profit %.

Algorithm:

Step 1: Start

Step 2: take inputs SP, CP

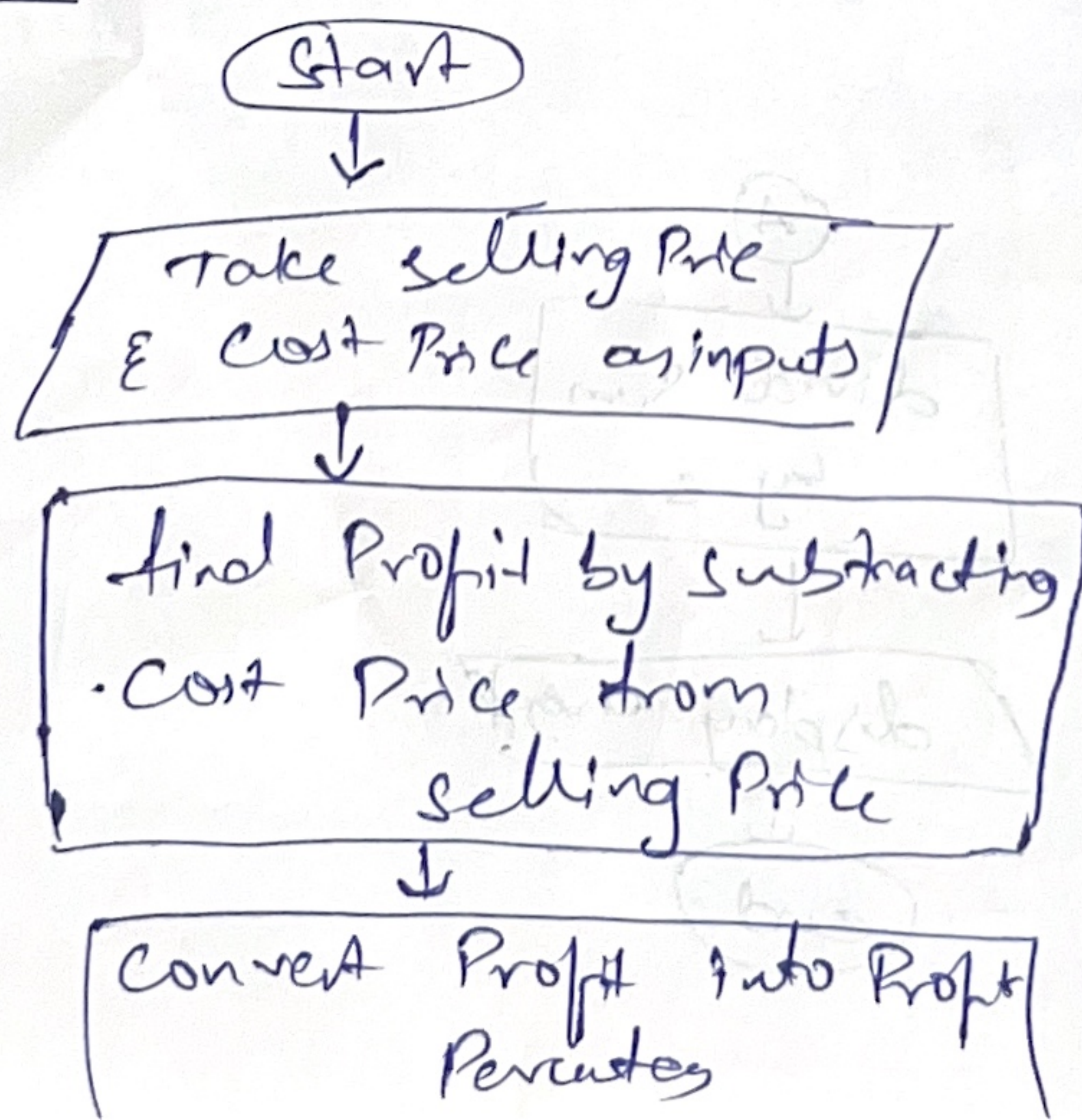
Step 3: find Profit by subtracting CP from SP

Step 4: Convert Profit into Profit Percentage by divide Profit by cost Price then multiply with 100 ($PP = \frac{P}{Cp} \times 100$)

Step 5: Display Profit Percentage

Step 6: End

Flowchart:



Display Profit Percentage

end

3. Find missing angle in triangle by taking 2 angles.

1. Input $\rightarrow$ 2 angles of triangle

2. Process $\rightarrow$ Add 2 angles, store in sum
subtract sum from 180

3. output $\rightarrow$ missing angle.

Algorithm:

1. Start

2. take two sides of Δ as input

3. Add 2 angle & store in a variable

4. Subtract sum from 180, store in

Variable.

5. display missing angle

6. End.

Flowchart:

Start

Take 2 angle as input

Add 2 angle & store in a variable sum

subtract sum from 180 & store in variable

display missing angle.

end

4. Find last digit of given number.

input = Take a number

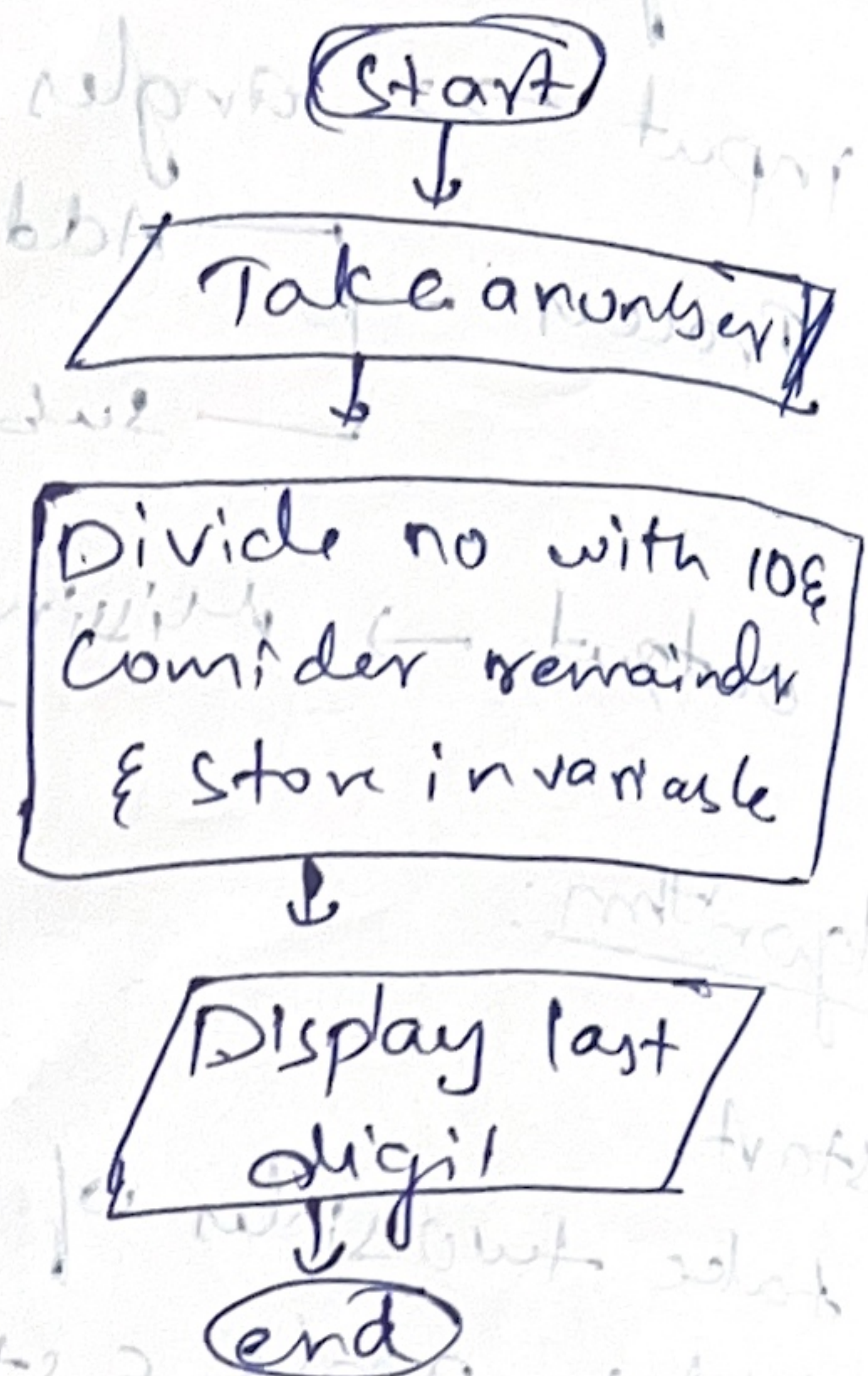
Process: — Divide no with 10

output: — consider remainder
last digit

algorithm.

1. Start
2. Take a number
3. Divide number with 10
& consider remainder
4. Store remainder in a
variable
5. Display last digit
6. End.

Flowchart



5. Remove last digit from given number.

Input: Take a number

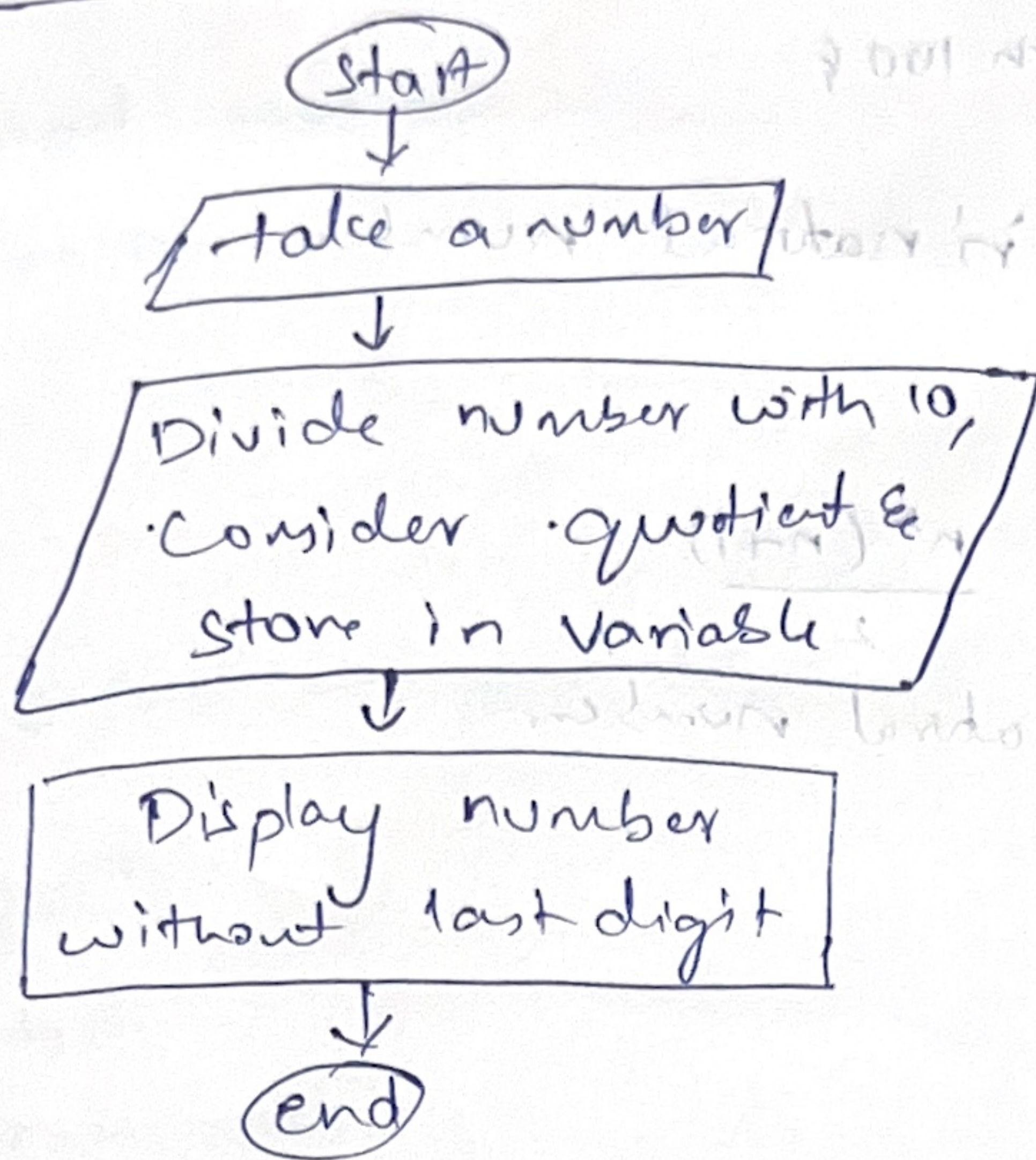
Process: Divide number with 10 &
consider quotient.

output: Number without last digit.

algorithm.

1. Start
2. Take a number
3. Divide number with 10 &
consider quotient & store in a variable
4. Display number without last digit
5. end

Flow chart



6. Find the first digit of a digit number.

Input: number (4 digit)

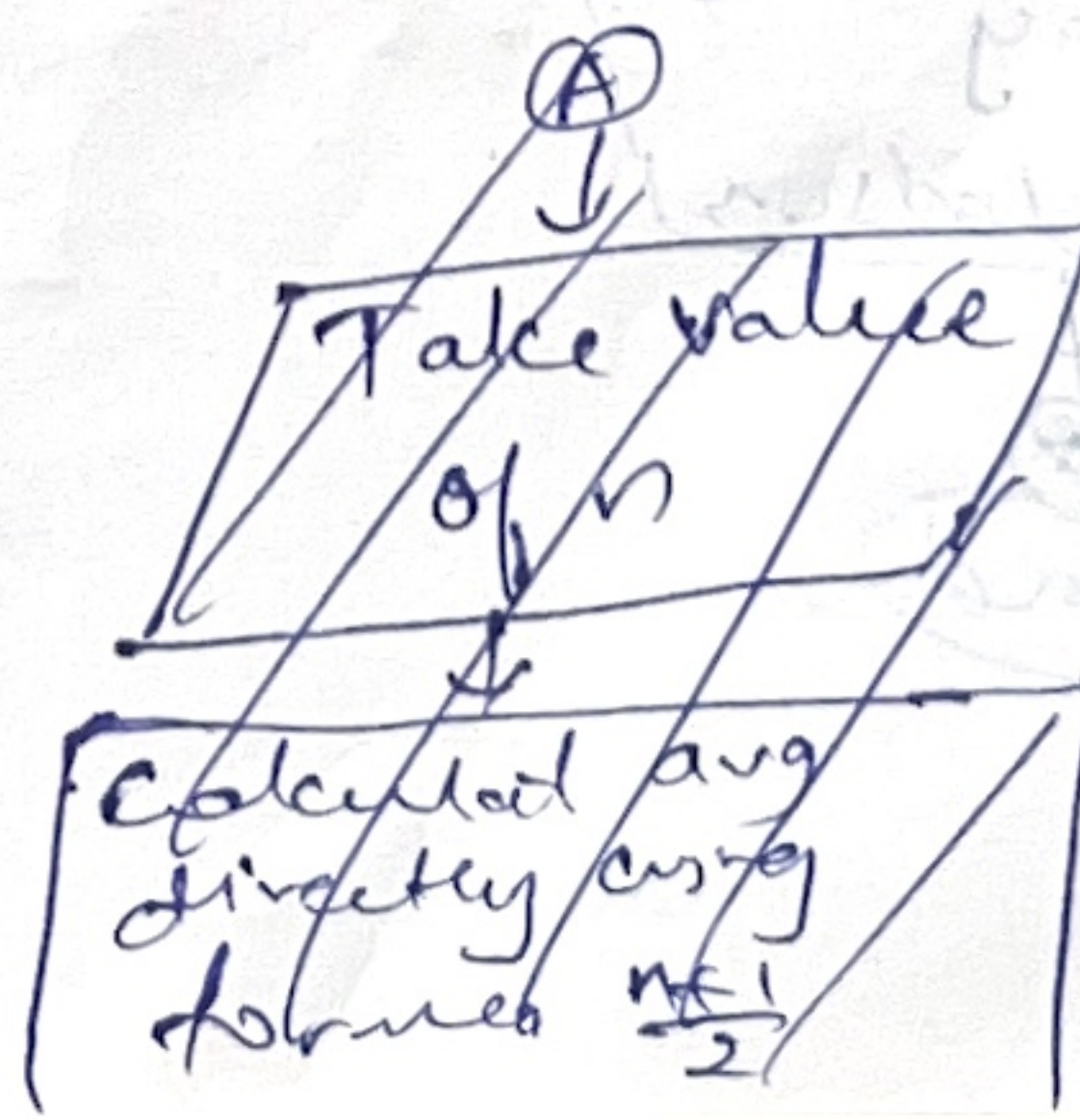
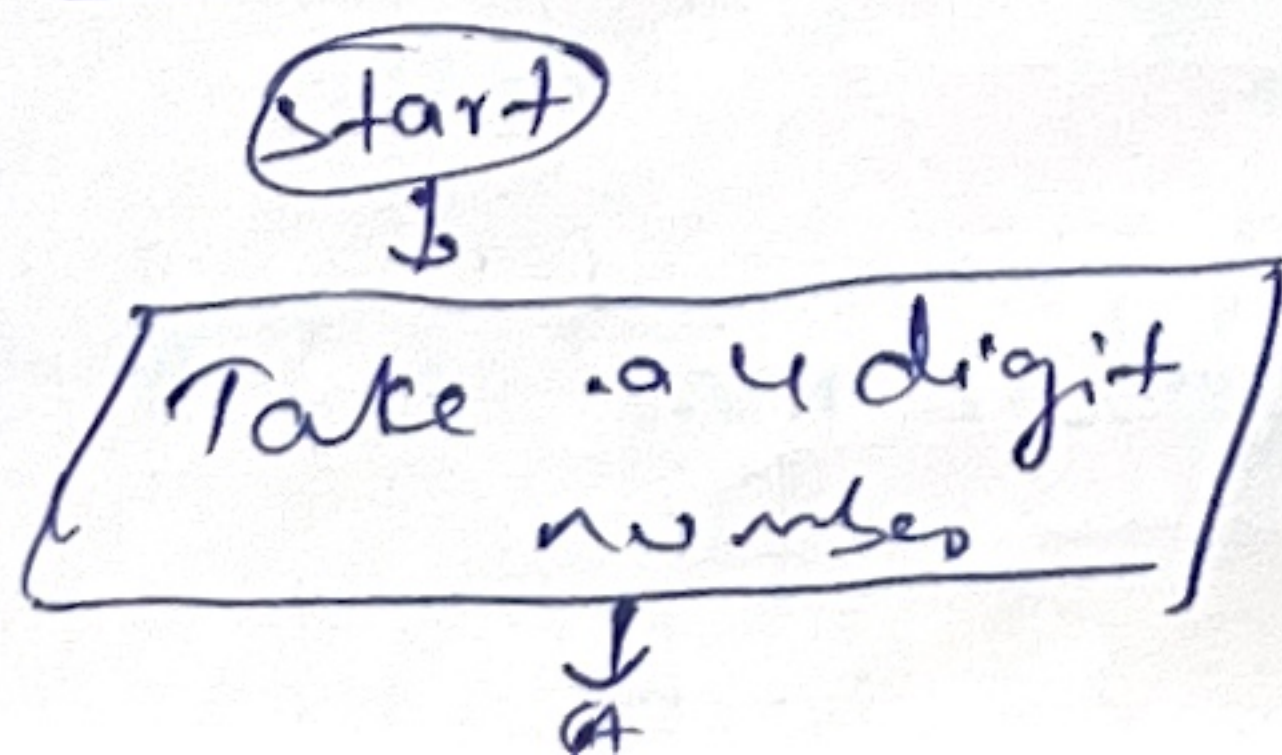
Process: Divide number with 1000 & consider quotient.

Output: first digit.

Algorithm

1. Start
2. Take a 4 digit number
3. Divide the number with 1000 & consider quotient & store in variable
4. Display first digit.
5. End.

Flowchart



Divide number with 100 &

⑦ Find Sum of first 'n' natural number.

input: number 'n'

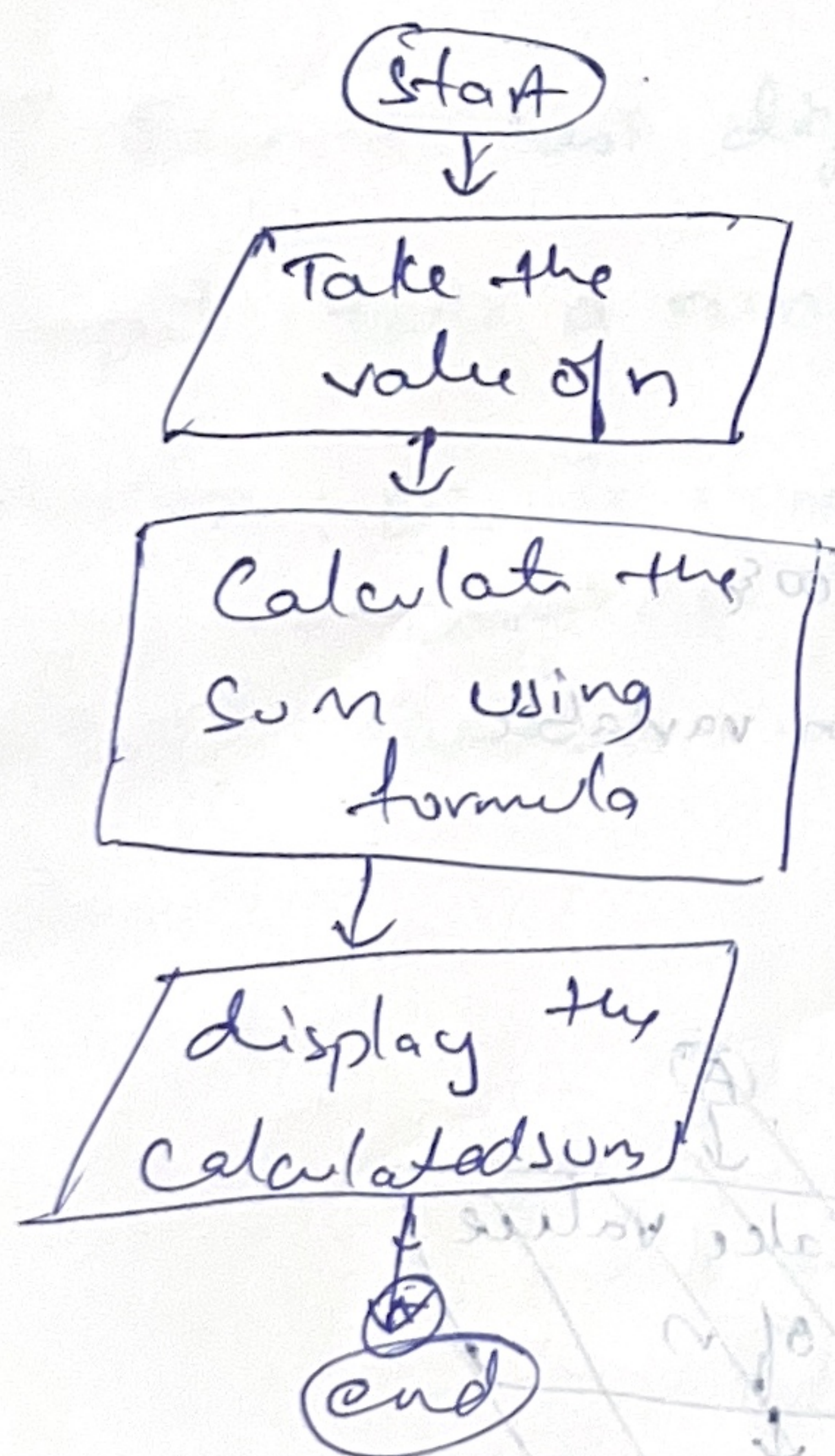
process: use formula $\frac{n \times (n+1)}{2}$

output: sum of n natural number.

Algorithm

1. Start
2. Take value of 'n'.
3. Calculate the sum using formula $\text{sum} = \frac{n(n+1)}{2}$
4. Display calculated sum.
5. End.

Flowchart



8. Find Avg of first n natural numbers.

input: number n

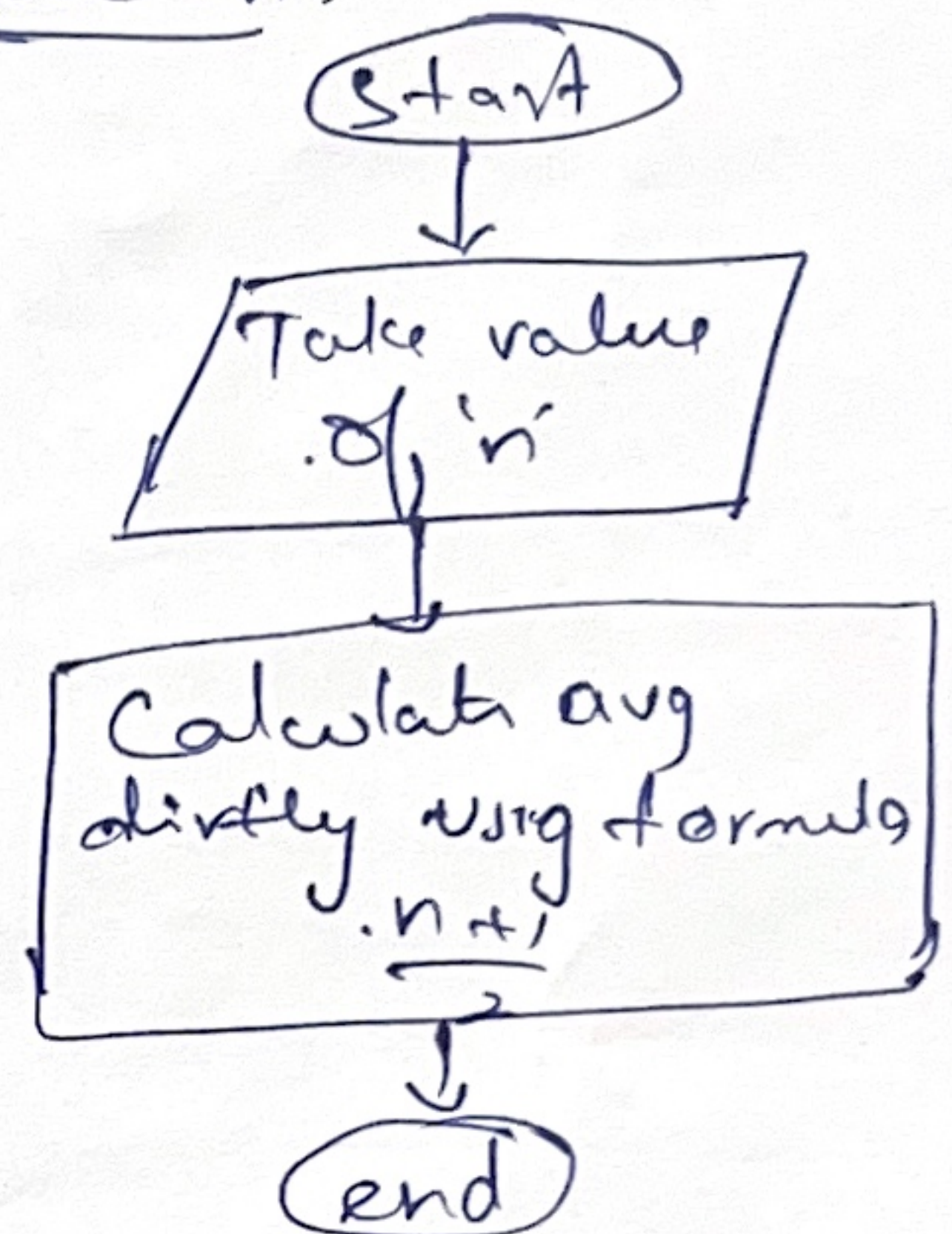
Process:
 - Find sum of n natural no's using $\frac{n(n+1)}{2}$
 - then divide sum with n
 i.e. sum/n

output: avg of first n natural no's.

Algorithm:

1. Start
2. Take value of n
3. Use formula $\frac{n+1}{2}$ & store in a variable
4. Display Calculated value
5. End.

Flowchart:



u. Swapping two numbers wrong

Ex $a = 10$
 $b = 20$

output

$a = 20$
 $b = 10$

$a = b$
 $b = a$

$a = 20$
 $a = b$

$b = a$

$a \rightarrow 10$
 $b \rightarrow 20$

a
 20

b
 20

with 3 variable

$$c = a$$

$$a = b$$

$$b = c$$

$$a = 10$$

$$b = 20$$

$$a = 10$$

$$b = 20$$

logic

$$c = a \rightarrow \boxed{10}$$

$$a = b \rightarrow \boxed{20}$$

$$b = c \rightarrow \boxed{10}$$

Print (a) = 20

Print (b) = 10

with 2 variable

$$a = 10$$

$$b = 20$$

logic

$$a = a + b$$

$$= 10 + 20$$

$$\boxed{a = 30}$$

$$b = a - b$$

$$= 30 - 20$$

$$\boxed{b = 10}$$

$$a = a - b$$

$$= 30 - 10$$

$$\boxed{a = 20}$$

$$a = a + b$$

$$b = a - b$$

$$a = a - b$$

o/p = a = 20
b = 10

$$\Rightarrow a = a * b \quad a = 10$$

$$b = a / b \quad b = 20$$

$$a = a / b$$

back ground

$$a = 200$$

$$b = \begin{cases} a = 10 \\ 200 \end{cases}$$

$$b = 10$$

$$b = \begin{cases} 20 \\ 10 \end{cases}$$

$$a = 20$$

$$b = 10$$

$$a = 20$$

$$a = \begin{cases} 200 \\ 20 \end{cases}$$

algorithm

1. Start
2. Take input a and b.
3. now multiply a and b and store in a, i.e. $a = a * b$
4. Next divide a and b store in b.
5. divide a and b store in a.
6. display a and b
7. exit

Flowchart

